Falak Pabari

69 Brown St, Box 1884 | Providence, RI 02912 | Phone: (401) 340-7593 | E-Mail: falak_pabari@brown.edu

EDUCATION

Brown University, *B.Sc. Computer Science and Applied Math*, 3.9/4.0 GPA Providence, RI | **Expected Graduation May 2026**

Relevant Courses**:** Computational Molecular Biology I & II, Intro to Cryptography & Cyber Security, Statistical Inference, Deep Learning in Genomics, Quantum Information, Mathematical Modeling of Infectious Diseases, Data Science

RESEARCH EXPERIENCE

Singh Lab, ***Undergraduate Research Assistant****, Brown University* Providence, RI | May 2025 – Present

- Applying deep learning to **DNA methylation and aging prediction**, developing VAEs and contrastive learning frameworks.
- Implemented **parallelized image generation and robust dataloading** for large-scale scatterplot datasets.
- Analyzed latent space representations with **PCA, t-SNE, cosine similarity heatmaps, and k-means clustering** to assess clustering quality.

Y. Ma Lab, ***Undergraduate Research Assistant****, Brown University* Providence, RI | May 2025 – Present

- Developing models for **single-cell perturbation analysis** (Mixscape, ContrastiveVI, SC-VAE) to isolate treatment-specific variation from background variation.
- Proposed a new method to **quantify uncertainty in perturbed cells** and benchmarked against existing methods.

Humans to Robots Lab, ***Undergraduate Research Assistant****, Brown University* Providence, RI | May 2023 – February 2025

- Collaborated to research human-robot interaction, focusing on behavioral modeling with real dogs and robot dogs.
- Lead efforts in data collection and analysis using Python and statistical methods to evaluate results.
- Contributed to the publication of the findings from the study of robot-dog interactions.

TEACHING EXPERIENCE

Multivariable Calculus (MATH 0200), *Brown University* Providence, RI | September – December 2024

- Held weekly problem-solving sessions, clarifying challenging concepts such as gradients, curl, and multiple integrals.
- Graded weekly homework for 40+ students, ensuring accuracy and providing constructive feedback.

Linear Algebra (MATH 0520), *Brown University* Providence, RI | September – December 2023

- Assisted in planning and leading office hours to help students understand vector spaces, eigenvalues, and diagonalization.
- Redesigned review materials to be more accessible for students, improving engagement and comprehension.

Object Oriented Programming (CSCI 0200), *Brown University* Providence, RI | September – December 2023

- Contributed to course improvements by developing new assignments, aligning them with real-world applications.
- Assisted students during office hours to debug code and strengthen problem-solving skills.

PROJECTS

SKIWI, *Co-founder* Providence, RI | March 2025 – Present

- Co-founded **SKIWI**, an AI-powered digital skincare platform that delivers **personalized, everyday skincare recommendations** using computer vision and dermatology-informed models.
- Created early-stage branding, website wireframes, and a marketing plan targeting students and young professionals.

Problem Solving Fellows, *Brown University* Providence, RI | September 2024 – Present

- Designed workshops and resources to support **study abroad, first-generation, and international students** in navigating new academic and cultural environments.
- Designed the official **Problem Solving Fellows program website** to improve visibility, streamline communication, and provide centralized resources for fellows. In process of launching

Advanced Predictive Analytics for RWQA in Kenya, *Data Scientist* Remote | May - September 2024

- Worked with an international team to develop predictive models for remote water quality assessment in rural areas of Kenya.
- Led efforts in data collection, processing, and modeling to optimize the prediction of water quality metrics using machine learning algorithms.
- Assisted in the deployment of the model for real-time assessments and contributed to drafting the project paper.

SKILLS & INTERESTS

Programming & Tools: Python, R, MATLAB, Git, Bash, LaTeX, Jupyter, scvi-tools, PyTorch, NumPy, pandas, scikit-learn, matplotlib, DESeq2, Figma

Data Science & ML: Variational Autoencoders, Contrastive Learning, PCA, t-SNE, Clustering, Differential Expression Analysis, Dynamic Programming

Domains: Computational Biology, Biostatistics, Cryptography, Applied Mathematics, Quantum Information